

Chapter 13. Double Integrals and Riemann Sums

Area accumulation, volume, numerical approximation, and iterated integration

Chapter goal. A single integral adds small contributions along a line. A double integral adds small contributions over a two-dimensional region. This chapter develops double integrals from Riemann sums, explains their geometric meaning, and shows how Fubini's theorem turns a two-dimensional accumulation problem into two ordinary one-variable integrations.

The ideas in this chapter are important bridges from calculus to applied mathematics, statistics, data science, and machine learning. A double integral may represent volume, mass, charge, expected value, average temperature, total risk, probability over a region, or total loss accumulated over a two-parameter domain.

$$\text{total amount} \approx \sum \text{value at a sample point} \times \text{small area}.$$

When the rectangles become very small, the approximation becomes an integral.

Learning objectives

After studying this chapter, you should be able to:

1. explain a double integral as the limit of double Riemann sums;
2. compute area elements $dA = \Delta x \Delta y$ on rectangular partitions;
3. interpret $\iint_R f(x, y) dA$ as volume when $f \geq 0$;
4. interpret double integrals as signed accumulation when f changes sign;
5. approximate double integrals using sample points and the midpoint rule;
6. evaluate double integrals over rectangles using Fubini's theorem;
7. compute average values over rectangles;
8. connect double integrals to density, mass, numerical integration, and probability.

1 From one-dimensional area to two-dimensional accumulation

Recall the definite integral of a continuous function $f(x)$ on an interval $[a, b]$:

$$\int_a^b f(x) dx = \lim_{\Delta x \rightarrow 0} \sum_i f(x_i^*) \Delta x.$$

Geometrically, if $f(x) \geq 0$, this is the area under the curve $y = f(x)$ above the interval $[a, b]$. Computationally, it means:

1. divide the interval into small subintervals;
2. choose a sample point in each subinterval;
3. multiply height by width;

4. add the pieces;
5. pass to the limit.

A double integral follows the same pattern, but the base region is two-dimensional. Instead of small intervals, we use small rectangles. Instead of length Δx , each small piece has area

$$\Delta A = \Delta x \Delta y.$$

If the height above (x, y) is $f(x, y)$, then a small rectangular column has approximate volume

$$f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Adding all such columns gives a double Riemann sum.

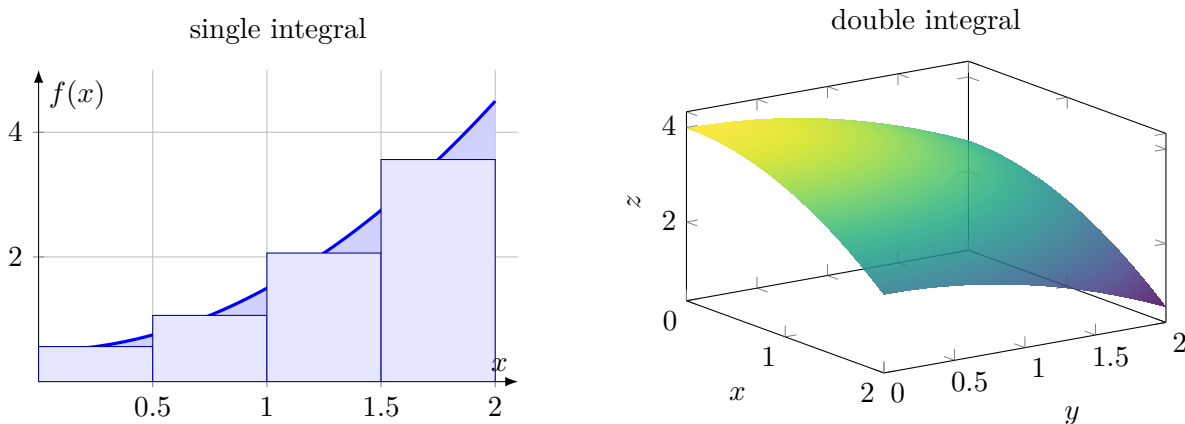


Figure 1: A single integral accumulates over an interval; a double integral accumulates over a region.

The main shift

A single integral adds over a line-like set. A double integral adds over an area-like set. The algebra looks similar, but the geometry changes from intervals to regions and from rectangles to columns.

2 Rectangular regions and partitions

For the first part of the theory, we focus on a rectangular region

$$R = [a, b] \times [c, d] = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}.$$

Divide $[a, b]$ into m equal subintervals and $[c, d]$ into n equal subintervals. Then

$$\Delta x = \frac{b-a}{m}, \quad \Delta y = \frac{d-c}{n}, \quad \Delta A = \Delta x \Delta y.$$

This divides R into mn small rectangles R_{ij} . In each R_{ij} , choose a sample point (x_{ij}^*, y_{ij}^*) . The associated double Riemann sum is

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

If $f(x, y) \geq 0$, this sum estimates the volume under $z = f(x, y)$ and above R .

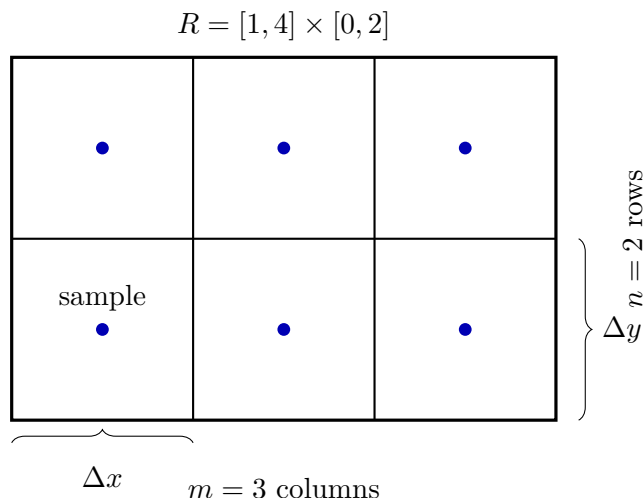


Figure 2: A rectangle divided into subrectangles. Each subrectangle contributes one term to a double Riemann sum.

3 The double integral on a rectangle

Let f be a function defined on a rectangle $R = [a, b] \times [c, d]$. If the double Riemann sums approach a single limiting value as the partition becomes finer, independent of the sample points, then f is integrable on R . The limit is called the *double integral* of f over R .

Definition: double integral over a rectangle

For a sufficiently nice function f on R ,

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

Here dA represents a small piece of area. In rectangular coordinates, it is approximated by $\Delta A = \Delta x \Delta y$.

The notation dA reminds us that the integral is adding over area. Later, when we change coordinates, the expression for dA will change. In rectangular coordinates,

$$dA = dx dy \quad \text{or} \quad dA = dy dx.$$

Volume interpretation

If $f(x, y) \geq 0$ on R , then

$$\iint_R f(x, y) dA$$

is the volume of the solid below the surface $z = f(x, y)$ and above the base region R .

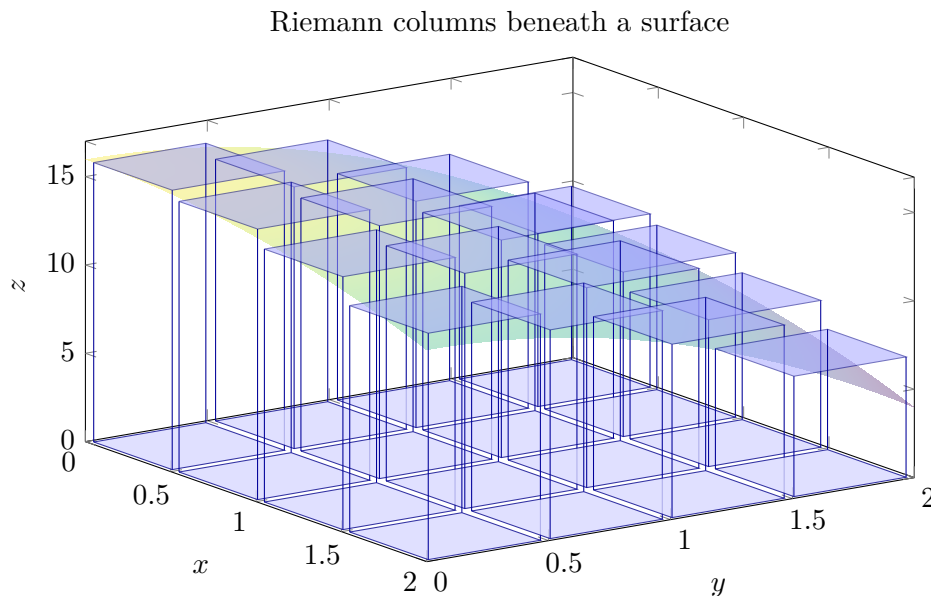


Figure 3: A double Riemann sum approximates volume by rectangular columns beneath $z = 16 - x^2 - 2y^2$.

4 Worked example: upper-right Riemann sum

Example 4.1: Upper-right sum for $f(x, y) = x^2y$

Consider

$$f(x, y) = x^2y, \quad R = [1, 4] \times [0, 2].$$

Use $m = 3$ subdivisions in the x -direction and $n = 2$ subdivisions in the y -direction. Then

$$\Delta x = \frac{4-1}{3} = 1, \quad \Delta y = \frac{2-0}{2} = 1, \quad \Delta A = 1.$$

Using upper-right corners as sample points gives

$$x = 2, 3, 4, \quad y = 1, 2.$$

Therefore

$$\begin{aligned} \sum_{i=1}^3 \sum_{j=1}^2 f(x_i^*, y_j^*) \Delta A &= f(2, 1) + f(3, 1) + f(4, 1) + f(2, 2) + f(3, 2) + f(4, 2) \\ &= 4 + 9 + 16 + 8 + 18 + 32 \\ &= 87. \end{aligned}$$

Since x^2y increases as x and y increase on this rectangle, upper-right corners overestimate the integral.

x	y	$f(x, y) = x^2y$	ΔA	contribution
2	1	4	1	4
3	1	9	1	9
4	1	16	1	16
2	2	8	1	8
3	2	18	1	18
4	2	32	1	32
Total				87

5 The midpoint rule for double integrals

The *midpoint rule* uses the center of each subrectangle as the sample point. For a rectangle divided into m by n equal subrectangles,

$$\iint_R f(x, y) dA \approx \sum_{i=1}^m \sum_{j=1}^n f(\bar{x}_i, \bar{y}_j) \Delta A,$$

where $\bar{x}_i$ and $\bar{y}_j$ are midpoint coordinates.

Example 5.1: Midpoint rule for x^2y

Again let

$$f(x, y) = x^2y, \quad R = [1, 4] \times [0, 2], \quad m = 3, \quad n = 2.$$

The midpoint sample values are

$$x = 1.5, 2.5, 3.5, \quad y = 0.5, 1.5.$$

Thus

$$\iint_R x^2y dA \approx \sum_{x \in \{1.5, 2.5, 3.5\}} \sum_{y \in \{0.5, 1.5\}} x^2y = 44.5.$$

The exact value, found later in this chapter, is 42, so the midpoint estimate is much closer than the upper-right estimate 87.

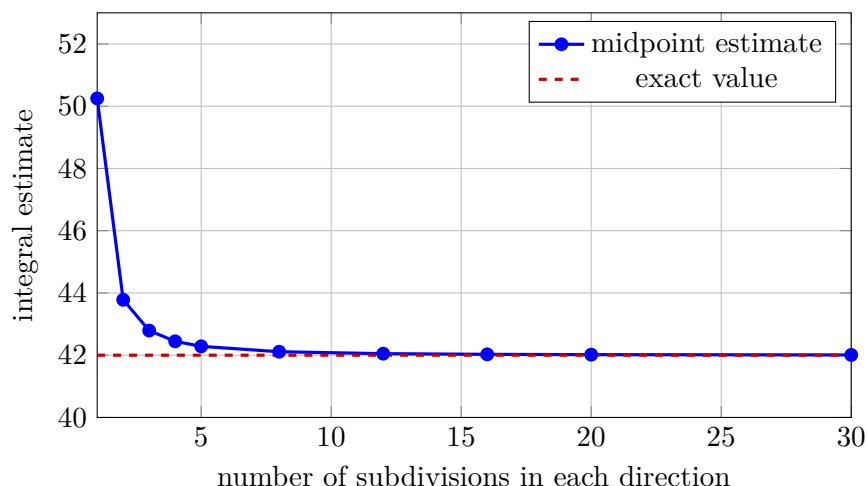


Figure 4: Midpoint Riemann sums converge to the exact double integral as the grid becomes finer.

Why midpoint sampling is often good

The midpoint rule samples each rectangle at its center. For smooth functions, this often balances overestimates and underestimates better than using only corners. This is one reason midpoint-type ideas appear in numerical integration, finite-volume methods, and simulation.

6 Fubini's theorem on rectangles

Riemann sums define the double integral, but limits of double sums are rarely the easiest way to compute exact answers. The key computational result is Fubini's theorem.

Fubini's theorem for rectangles

If f is continuous on $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

The double integral can be computed by integrating one variable at a time.

The inner integral treats the other variable as constant. For example,

$$\int_1^4 x^2 y dx = y \int_1^4 x^2 dx = y \left[\frac{x^3}{3} \right]_1^4 = 21y.$$

Then

$$\int_0^2 21y dy = \left[\frac{21y^2}{2} \right]_0^2 = 42.$$

Therefore

$$\iint_{[1,4] \times [0,2]} x^2 y dA = 42.$$

The same answer is obtained in the other order:

$$\int_1^4 \int_0^2 x^2 y \, dy \, dx = \int_1^4 2x^2 \, dx = 42.$$

The order changes the intermediate expressions, but not the final value.

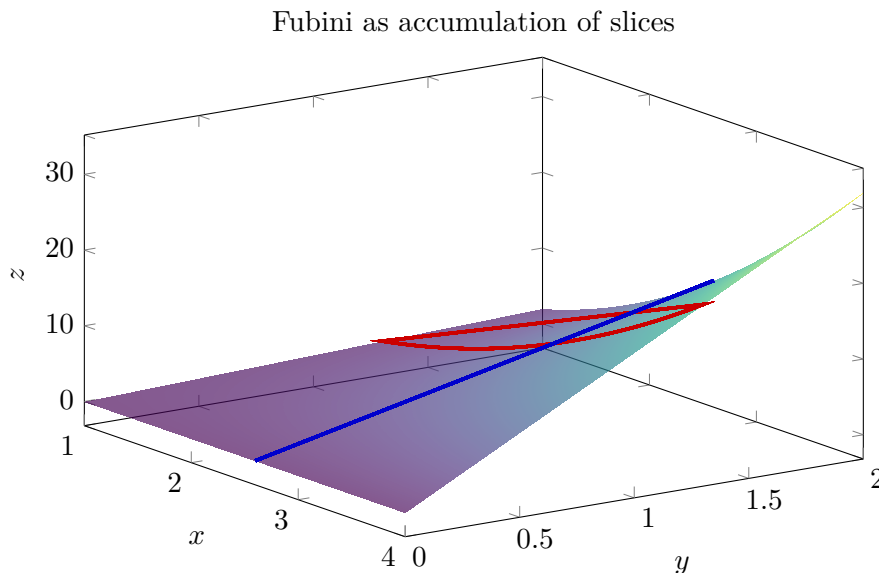


Figure 5: Fubini's theorem computes a double integral by accumulating one-dimensional slices of the surface.

7 Separability and product structure

Sometimes a function separates into a product of a function of x and a function of y :

$$f(x, y) = g(x)h(y).$$

On a rectangle, Fubini's theorem gives

$$\iint_{[a,b] \times [c,d]} g(x)h(y) \, dA = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right).$$

For example,

$$\iint_{[0,\pi] \times [0,1]} y \sin x \, dA = \left(\int_0^\pi \sin x \, dx \right) \left(\int_0^1 y \, dy \right) = 2 \cdot \frac{1}{2} = 1.$$

This product structure appears frequently in probability. If two random variables are independent, their joint density often factors as $p(x, y) = p_X(x)p_Y(y)$.

8 Signed accumulation

If $f(x, y)$ takes both positive and negative values, then

$$\iint_R f(x, y) \, dA$$

is no longer ordinary volume. It is *signed volume*: positive parts above the xy -plane count positively, and negative parts below the xy -plane count negatively. For example, on a symmetric rectangle,

$$\iint_{[-1,1] \times [0,2]} xy \, dA = 0,$$

because the positive and negative contributions cancel.

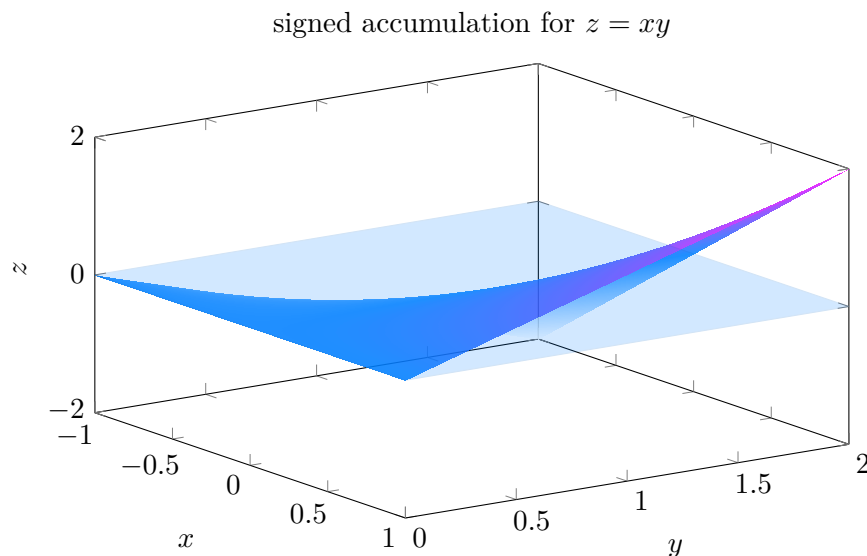


Figure 6: A double integral can measure signed accumulation when the surface has positive and negative parts.

Volume versus signed volume

If $f \geq 0$, the double integral is a volume. If f changes sign, the double integral is signed volume. To compute total physical volume between the surface and the plane, one would integrate $|f(x, y)|$ instead.

9 Average value over a rectangle

The average value of a one-variable function over $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

The average value of a two-variable function over a region R is

$$f_{\text{avg}} = \frac{1}{\text{Area}(R)} \iint_R f(x, y) \, dA.$$

For a rectangle $R = [a, b] \times [c, d]$,

$$\text{Area}(R) = (b-a)(d-c).$$

Example 9.1: Average height

Let

$$f(x, y) = 16 - x^2 - 2y^2, \quad R = [0, 2] \times [0, 2].$$

The double integral is

$$\iint_R (16 - x^2 - 2y^2) dA = \int_0^2 \int_0^2 (16 - x^2 - 2y^2) dy dx = \frac{136}{3}.$$

Since $\text{Area}(R) = 4$, the average value is

$$f_{\text{avg}} = \frac{1}{4} \cdot \frac{136}{3} = \frac{34}{3}.$$

In a physical interpretation, this is the average height of the surface over the rectangular base.

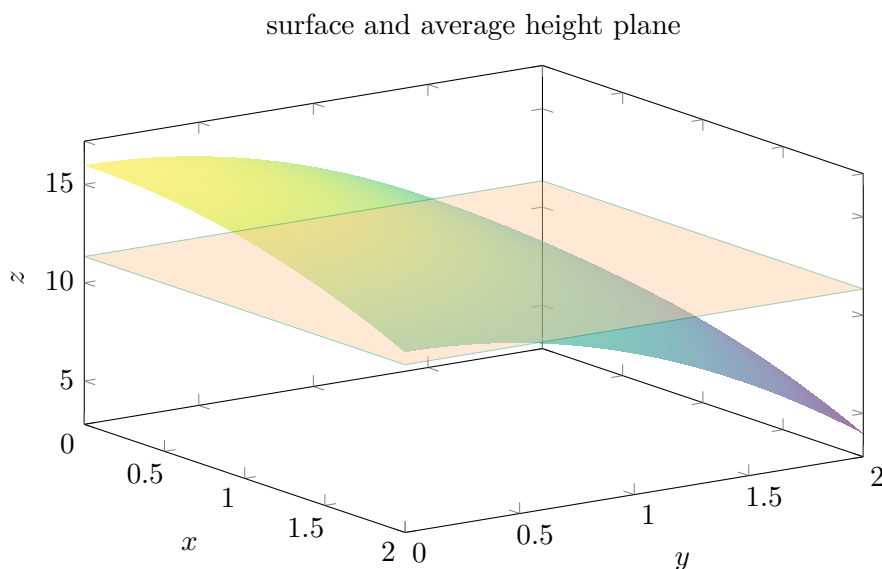


Figure 7: The average value is the constant height that gives the same total volume over the base region.

10 Mass and density on a rectangular plate

Suppose a thin rectangular plate occupies a region R in the xy -plane. If its density at (x, y) is $\rho(x, y)$, then the mass of a small rectangle is approximately

$$\rho(x_{ij}^*, y_{ij}^*) \Delta A.$$

The total mass is therefore

$$M = \iint_R \rho(x, y) dA.$$

Example 10.1: Mass of a rectangular plate

Let

$$\rho(x, y) = 2 + x + y, \quad R = [0, 3] \times [0, 2].$$

Then

$$\begin{aligned} M &= \int_0^3 \int_0^2 (2 + x + y) \, dy \, dx \\ &= \int_0^3 \left[2y + xy + \frac{y^2}{2} \right]_0^2 \, dx \\ &= \int_0^3 (6 + 2x) \, dx = 27. \end{aligned}$$

The double integral is not just a geometric volume. It is a general accumulation tool: density times area gives mass, and the integral adds these contributions.

density $\rho(x, y) = 2 + x + y$ on a plate

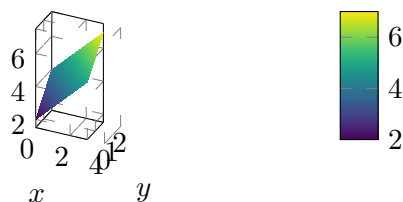


Figure 8: A density field over a rectangular plate. Integrating density over area gives mass.

11 Numerical double integration

In applications, exact symbolic integration may not be available. A function may be known only through data, simulation, or a black-box model. Then Riemann sums become algorithms.

A reusable midpoint rule for rectangles has the form

$$\text{estimate} = \sum_{i=1}^m \sum_{j=1}^n f(\bar{x}_i, \bar{y}_j) \Delta x \Delta y.$$

For high-dimensional applied mathematics and machine learning, this point of view matters. Integrals are often approximated numerically, and sums over grids, samples, or data points are discrete versions of continuous accumulation.

12 Monte Carlo viewpoint

A double integral over a rectangle can also be estimated by random sampling. If R has area A and (X, Y) is sampled uniformly from R , then

$$\iint_R f(x, y) \, dA = A \mathbb{E}[f(X, Y)].$$

This gives the Monte Carlo estimator

$$\iint_R f(x, y) dA \approx A \cdot \frac{1}{N} \sum_{k=1}^N f(X_k, Y_k).$$

Monte Carlo integration is usually less accurate than grid-based integration in low dimensions, but it becomes central in statistics, Bayesian computation, uncertainty quantification, and high-dimensional integration.

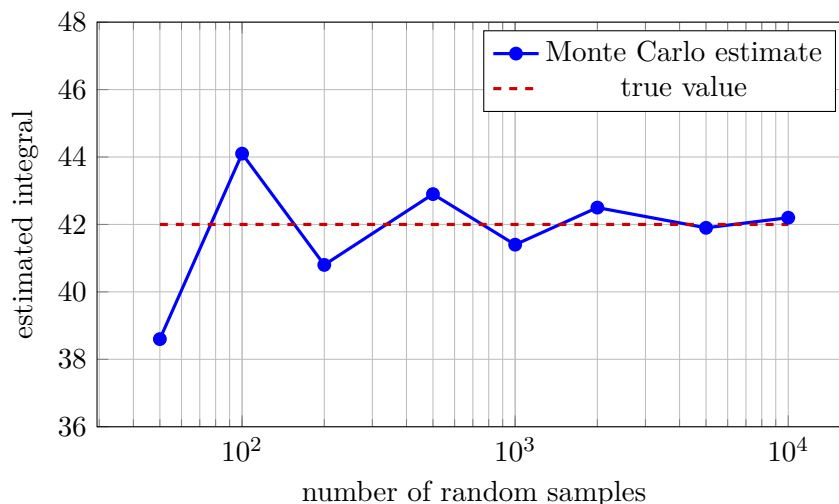


Figure 9: Monte Carlo estimates fluctuate randomly but approach the true integral as the number of samples increases.

13 Double integrals and probability on rectangles

If $p(x, y)$ is a joint probability density on a region R , then probabilities are double integrals:

$$P((X, Y) \in A) = \iint_A p(x, y) dA.$$

For a rectangular event $A = [a, b] \times [c, d]$, we compute

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d p(x, y) dy dx.$$

This is one reason double integrals are essential preparation for statistics. Later chapters will develop non-rectangular regions, change of variables, and multivariate normal densities in more detail.

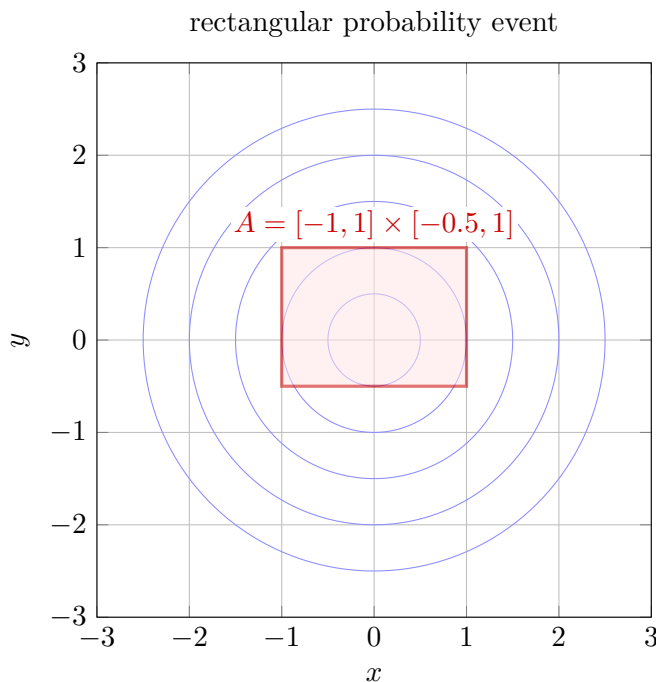


Figure 10: Probability over a rectangle is the double integral of a density over that rectangle.

14 Common mistakes

Mistake 1: forgetting ΔA

A Riemann sum must have the form

$$\sum f(x_{ij}^*, y_{ij}^*) \Delta A.$$

The values of f alone do not approximate an integral. They must be multiplied by area.

Mistake 2: confusing the order of integration

The expression

$$\int_a^b \int_c^d f(x, y) dy dx$$

means: integrate with respect to y first, then integrate the resulting function of x with respect to x .

Mistake 3: assuming every double integral is volume

A double integral is ordinary volume only when $f \geq 0$. Otherwise it is signed accumulation.

Mistake 4: using Fubini without checking the region

In this chapter, the main region is a rectangle. In the next chapter, we will study general regions where bounds may depend on variables.

15 Chapter exercises**Conceptual problems**

1. Explain in words why the term $f(x_{ij}^*, y_{ij}^*)\Delta A$ represents a small volume when $f \geq 0$.
2. What is the difference between a double Riemann sum and a double integral?
3. Why does dA represent area rather than length?
4. Explain why $\iint_R f(x, y) dA$ may be negative if f changes sign.
5. Describe the difference between a midpoint estimate and an upper-right corner estimate.

Computational problems

6. Let $f(x, y) = 3x + 2y$ on $R = [0, 2] \times [1, 3]$. Use $m = 2$, $n = 2$, and upper-right sample points to approximate $\iint_R f dA$.
7. Repeat Problem 6 using midpoint sample points.
8. Compute the exact value of $\iint_R (3x + 2y) dA$.
9. Evaluate $\iint_{[0,1] \times [0,2]} xy^2 dA$.
10. Evaluate $\iint_{[1,3] \times [0,\pi]} x \cos y dA$.
11. Evaluate $\iint_{[-1,1] \times [-1,1]} x^3 y^2 dA$ without doing a long calculation. Explain the cancellation.
12. Find the average value of $f(x, y) = x + y$ on $[0, 4] \times [0, 2]$.
13. Find the mass of a rectangular plate $[0, 2] \times [0, 3]$ with density $\rho(x, y) = 1 + x + 2y$.
14. Find the volume below $z = 9 - x^2 - y^2$ and above $[0, 1] \times [0, 2]$.
15. Compute $\iint_{[0,1] \times [0,1]} e^{x+y} dA$.

Modeling and interpretation problems

16. A temperature field on a rectangular plate is $T(x, y) = 20 + 3x - y$. Write a double integral for the average temperature over $[0, 4] \times [0, 2]$ and evaluate it.
17. A risk surface over two model parameters is $L(\theta_1, \theta_2) = \theta_1^2 + 4\theta_2^2$. Compute the average risk over $[-1, 1] \times [-1, 1]$.
18. Suppose $p(x, y) = cxy$ on $[0, 1] \times [0, 1]$. Find c so that p is a probability density.
19. For the density in Problem 18, compute $P(0 \leq X \leq 1/2, 0 \leq Y \leq 1)$.

20. Explain why numerical integration is important when $f(x, y)$ is only available through simulation or data.

Computational lab problems

21. Write a midpoint-rule function to approximate $\iint_{[0,2] \times [0,2]} (16 - x^2 - 2y^2) dA$.
22. Compare midpoint estimates for several subdivisions in each direction.
23. Use random sampling to estimate $\iint_{[1,4] \times [0,2]} x^2 y dA$.
24. Plot the error of the midpoint rule as the number of subdivisions increases.
25. Plot the surface $z = x^2 y$ above $[1, 4] \times [0, 2]$ and add a coarse grid of columns.

16 Python lab and interactive page

- Notebook lab: `../labs/Chapter13_Double_Integrals_and_Riemann_Sums.ipynb`
- Interactive HTML page: `../htmls/chapter-13.html`

17 Chapter summary

A double integral is the limit of double Riemann sums. On a rectangle, the approximation

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

becomes the exact quantity

$$\iint_R f(x, y) dA.$$

When $f \geq 0$, the integral is volume. When f changes sign, it is signed accumulation. When f is a density, it computes mass or probability. Fubini's theorem makes double integrals computable by reducing them to iterated one-variable integrals:

$$\iint_{[a,b] \times [c,d]} f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

This chapter prepares the next step: double integrals over non-rectangular regions, where the bounds themselves encode geometry.